

Project Proposal

Project Title: Disease Risk Prediction System

Team Members: Amanpreet Bhatia, Mansi Gaur

Motivation / Problem Statement

Early disease detection is critical for improving patient outcomes and reducing long-term healthcare costs. Traditional risk assessment procedures can be time-intensive and subject to variability across practitioners. This project proposes a machine learning based disease risk prediction system that leverages patient health data to identify individuals at elevated risk and support proactive clinical decision-making. The system is designed to be disease-agnostic, enabling adaptation across multiple conditions. As a proof of concept, heart disease prediction will be demonstrated in the final presentation.

Objectives & Research Questions

Objectives

- Develop a generalized machine learning framework for disease risk prediction using patient health indicators
- Compare multiple classification algorithms to assess trade-offs between predictive performance and interpretability
- Design an interpretable prediction workflow that highlights key features contributing to model decisions
- Apply robust data preprocessing, validation, and ethical data handling practices
- Demonstrate system extensibility through a heart disease prediction use case

Research Questions

- How effectively can machine learning models predict disease risk based on patient health indicators?
- Which ML models provide the best balance between performance and interpretability in a healthcare context?
- Which clinical features strongly influence disease risk predictions across conditions?
- What limitations and ethical considerations arise in applying ML to disease risk prediction?

Proposed Methods / Models

- Data Processing & Analysis - Pandas, NumPy
- Modeling - Scikit-learn, with optional XGBoost for ensemble methods

- Evaluation - Accuracy, precision, recall, F1-score, cross-validation
- Visualization - Matplotlib, Seaborn
- Explainability - SHAP for model-agnostic interpretability and clinical insight

Approach - The project prioritizes interpretable models, beginning with simpler algorithms and introducing more complex models if performance improvements justify the trade-off. Emphasis is placed on reliable evaluation and generalization.

Data / Materials

Open-source datasets will be used to support reproducibility and ethical data use. The primary dataset will be sourced from the UCI Machine Learning Repository, with heart disease data serving as the initial implementation and demonstration case. Additional datasets (e.g., diabetes or stroke) may be explored to assess the system's adaptability to multiple diseases.

Expected Outcomes

- A generalized, reusable disease risk prediction framework implemented in Python
- A reproducible, well-documented pipeline covering data preprocessing, modeling, and evaluation
- A functional prediction workflow allowing users to input patient data and receive interpretable risk predictions
- Comprehensive documentation covering assumptions, limitations, and ethical considerations

Timeline

- Week 1: Scope finalization, dataset selection, initial data inspection
- Week 2: Exploratory Data Analysis (EDA)
- Week 3: Data preprocessing and baseline model development
- Week 4: Tree-based models and hyperparameter tuning
- Week 5: Advanced modeling (XGBoost, optional) and comparison
- Week 6: Error analysis and model explainability
- Week 7: Prediction workflow development and documentation
- Week 8: Final report, visualization, and presentation preparation